

Over-operating free-air temperature range (unless otherwise noted). See [§ 9.1.8](#) for more information on cell configurations. All voltages are relative to VSS, except "Cell input differential."

	MIN	TYP	MAX	UNIT
T _{OPR} Operating free-air temperature	- 40		85	°C

7.4 Thermal Information

Over-operating free-air temperature range (unless otherwise noted)

THERMAL METRIC ⁽¹⁾	TSSOP			UNIT
	BQ76920xy 20 PINS (PW)	BQ76930xy 30 PINS (DBT)	BQ76940xy 44 PINS (DBT)	
R _{θJA, High K} Junction-to-ambient thermal resistance	93.7	86.5	70.1	°C/W
R _{θJC(top)} Junction-to-case(top) thermal resistance	28.7	19.4	17.5	°C/W
R _{θJB} Junction-to-board thermal resistance	44.6	41.3	33.9	°C/W
ψ _{JT} Junction-to-top characterization parameter	1.3	0.5	0.5	°C/W
ψ _{JB} Junction-to-board characterization parameter	44.1	40.6	33.4	°C/W
R _{θJC(bottom)} Junction-to-case(bottom) thermal resistance	n/a	n/a	n/a	°C/W

(1) For more information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics Application Report \(SPRA953\)](#).

7.5 Electrical Characteristics

Typical conditions are measured at 25°C with nominal BAT voltages of 18 V (BQ76920), 36 V (BQ76930), or 48 V (BQ76940) with V_{CELL} = 4 V. Min and max values include full recommended operating condition temperature range from - 40°C to +85°C. Certain characteristics may be shown at different voltage or temperature ranges, as clarified in the Test Condition sections.

PARAMETER		TEST CONDITION	MIN	TYP	MAX	UNIT
SUPPLY CURRENTS						
I _{DD}	NORMAL mode: ADC off, CC off	Sum of ICC_BAT and ICC_REGSRC currents		40	60	μA
	NORMAL mode: ADC on, CC off			60	90	
	NORMAL mode: ADC off, CC on			110	165	
	NORMAL mode: ADC on, CC on			130	195	
I _{CC_BAT}	NORMAL mode: ADC off	Into BAT pin		30	45	
	NORMAL mode: ADC on			50	75	
I _{CC_REGSRC}	NORMAL mode: CC off	Into REGSRC pin		10	15	
	NORMAL mode: CC on			80	120	
I _{SHIP}	SHIP/SHUTDOWN mode	Device in full shutdown, only VSTUP/BG and BOOT detector on		0.6	1.8	
LEAKAGE AND OFFSET CURRENTS						
dI _{NOM}	NORMAL mode supply current offset	Measured into VC5x (BQ76930, BQ76940) and VC10x (BQ76940)	- 5	±2.5	5	μA
dI _{SHIP}	SHIP mode supply current offset		- 1.0	±0.1	1.0	
dI _{ALERT}	Supply current when ALERT active	Measured into VC5x (BQ76930, BQ76940) or added to BAT (BQ76920)		15	25	
dI _{CELL}	Cell measurement input current	Measured into VC0 - VC15 except VC5, VC10, VC15	- 0.3	±0.1	0.3	
		Measured into VC5, VC10, VC15			0.5	
I _{LKG}	Terminal input leakage				1	